University of California, Davis
Department of Economics

PRELIMINARY EXAMINATION FOR THE Ph.D. DEGREE

MACROECONOMICS

July 2nd, 2024

Directions: The exam consists of five questions. Questions 1,2 concern ECN 200D, question 3 concerns ECN 200E, and questions 4,5 concern ECN 200F. Answer four out of the five questions. If you prefer (and have time), you can answer all five questions, and your grade will be based upon the best four scores. Feel free to impose additional structure on the problems below, but please state your assumptions clearly. You have 5 hours to complete the exam and an additional 20 minutes of reading time.

Question 1 (*25 points*)

Consider the standard neoclassical growth model in discrete time. There is a large number of identical households normalized to 1. Each household wants to maximize life-time discounted utility

$$U(\{c_t\}_{t=0}^{\infty}) = \sum_{t=0}^{\infty} \beta^t u(c_t), \ \ \beta \in (0,1).$$

Each household has an initial capital k_0 at time 0, and one unit of productive time in each period that can be devoted to work. Final output is produced using capital and labor, according to a CRS production function F. This technology is owned by firms (whose measure does not really matter because of the CRS assumption). Output can be consumed (c_t) or invested (i_t). Households own the capital (so they make the investment decision), and they rent it out to firms. Let $\delta \in [0,1]$ denote the depreciation rate of capital. Households own the firms, i.e., they are claimants to the firms' profits, but these profits will be zero in equilibrium.

The function u is twice continuously differentiable and bounded, with $u'(c) > 0$, $u''(c) < 0$, $u'(0) = \infty$, and $u'(\infty) = 0$. Regarding the production technology, we will introduce the useful function $f(x) \equiv F(x,1) + (1-\delta)x$, $\forall x \in \mathbb{R}_+$. The function f is twice continuously differentiable with $f'(x) > 0$, $f''(x) < 0$, $f(0) = 0$, $f'(0) = \infty$, and $f'(\infty) = 1 - \delta$.

In this model the government taxes households' **investment** at the constant rate $\tau \in [0,1]$. The government returns all the tax revenues, T, to the households in the form of lump-sum transfers. Throughout this question focus on recursive competitive equilibrium (RCE).

a) Write down the problem of the household recursively.[1] Carefully distinguish between aggregate and individual state variables. Then, define a RCE. (I recommend that you denote the aggregate capital of the economy as $\bar{k}$. That way I can clearly distinguish it from the individual state k in your scripts. Please do **not** use K for the aggregate state, as it is much harder to distinguish in hand-written notes.)

b) Write down the dynamic equation that the aggregate capital stock follows in this economy.

c) Now focus on steady-states. Describe the steady-state equilibrium value of the aggregate capital stock in this economy, and denote it by $\bar{k}^*(\tau)$.

d) Provide a closed-form solution for $\bar{k}^*(\tau)$ in the case of a Cobb-Douglas production function, $F(\bar{k}, n) = A\bar{k}^a n^{1-a}$, $a \in (0,1)$.

[1] Here firms face a static problem. I am not asking you to explicitly spell it out, but it will be critical for correctly defining a RCE.

For the rest of this question assume there is **no government**. In class, we often derived closed-form solutions for the value function (VF) and the policy function of our dynamic problems using the so-called "guess and verify" technique, but we did so within the context of a social planner's problem. Here we will explore this technique for our RCE specification. To make things simpler we will continue assuming a Cobb-Douglas production function, $F(\bar{k}, n) = A\bar{k}^a n^{1-a}$, we will also assume that $u(c) = ln(c)$ (the natural logarithm), and we will set $\delta = 1$. (Taken together, these three simplifications mean that we are in the well-known Brock-Mirman case.)

e) If you were to use the guess and verify method to derive the VF, $V(k, \bar{k})$, of the individual household who takes the aggregate law of motion of capital as given, what would that guess be? To be perfectly clear on what I am asking: in class we guessed that the planner's VF has the form $V(k) = \gamma_0 + \gamma_1 ln(k)$, where γ_0, γ_1 are unknowns to be determined. Here I want you to do the same, i.e., I want you to guess the *form* of the household's VF, which will depend on some parameters (you can call them γ's, like above), and, of course, on the state variables $(k, \bar{k})$. I do **not** want you to calculate these unknown parameters; only to make a correct guess.[2]

f) What is the form of the policy function of the individual household, $k' = g(k, \bar{k})$? To answer this use your guess for the VF from part (e) into the household's Bellman equation, obtain the first-order condition with respect to k', and solve for the optimal decision k'.[3]

[2] **Hint:** Since the household's VF has two state variables, we need a new and educated guess. To that end, recall the method of successive iterations from class. Define the sequence $\{V_n(k, \bar{k})\}_0^\infty$ as

$$V_{n+1}(k, \bar{k}) = \max_{k'} \Big\{ ln(c) + \beta V_n(k', \bar{k}') \Big\},$$

where c satisfies the household's budget constraint and $\bar{k}'$ follows the aggregate law of motion of capital. It turns out that after **just one** iteration the VF will already admit the form of the true solution (i.e., after just one iteration you will "see" the shape/form of V appearing)! All you need to do is start from a clever initial condition $V_0(k, \bar{k})$, apply only one iteration, and the resulting $V_1(k, \bar{k})$ will point you to the correct guess. Then the question is, what is a clever choice for $V_0(k, \bar{k})$? The Contraction Mapping Theorem tells us that it does not matter if the initial guess is bad. Take advantage of this fact, and choose the *easiest possible* V_0, even if it looks completely wrong!

[3] When you first solve for the optimal k' you will see that it also depends on $\bar{k}'$. To get rid of this term, guess that the aggregate law of motion of capital has the form $\bar{k}' = sF(\bar{k}, 1) = sA\bar{k}^a$, $s \in (0, 1)$. After imposing this (correct) conjecture, the policy function should depend only on (some of) the γ parameters from part (e), the newly introduced parameter s, and the two states, $(k, \bar{k})$.

Question 2 (*25 points*)

This question studies the co-existence of money and credit. Time is discrete with an infinite horizon. Each period consists of two subperiods. In the day, trade is bilateral and *partially* anonymous as in Kiyotaki and Wright (1991) (call this the KW market). At night trade takes place in a Walrasian or centralized market (call this the CM). There are two types of agents, buyers and sellers, and the measure of both is normalized to 1. The per period utility for buyers is $u(q) + X - H$, and for sellers it is $-q + X - H$, where q is the quantity of the special good produced by the seller and consumed by the buyer, X is consumption of the CM good (the numeraire), and H is hours worked in the CM. In the CM, all agents can turn one unit of labor into one unit of good. The function u satisfies $u' > 0$ and $u'' < 0$. Define the first-best quantity traded in the KW market as $q^* \equiv \{q : u'(q^*) = 1\}$.

In this model there are two types of sellers. Type-1 sellers, with measure $1 - \sigma$, never accept unsecured credit. Hence, in any meeting with this type of seller, the buyer must pay on the spot (*quid pro quo*) with money. In contrast, type-2 sellers, with measure σ, accept money but they also accept unsecured credit, in the form of a promise by the buyer to repay the seller in the **forthcoming CM with numeraire good**. However, there is a credit limit: the buyer can credibly promise to repay only an amount up to $C < q^*$.

A few more assumptions to close the model. Goods are non storable. There exits a storable and recognizable object, fiat money, that can serve as a medium of exchange. Money supply is controlled by a monetary authority, and we consider simple policies of the form $M_{t+1} = (1 + \mu)M_t$, $\mu > \beta - 1$. New money is introduced, or withdrawn if $\mu < 0$, via lump-sum transfers to buyers in the CM. Let ϕ denote the unit price of money (in terms of the numeraire). For simplicity, let us assume that in KW meetings buyers have all the bargaining power.

a) Describe the CM value function of the typical buyer. (Be sure to correctly identify the state variables; here there is a new state compared to the baseline model.)

b) Let q_j denote the quantity of good and d_j the amount of money exchanged in a meeting of type-$j = 1, 2$. Describe the bargaining solution in a typical type-j meeting.[4]

c) Describe the objective function of the representative buyer, $J(m')$.[5]

[4] **Hint:** Feel free to analyze the bargaining problems in detail, but I think that with a little intuitive thinking you can answer this question in a couple of minutes. In particular, you should think how the amount of money exchanged in a type-2 meeting is affected by the exogenous credit limit C.

[5] **Hint:** If it helps, you should feel free to skip all the formal work and obtain J based on your intuitive understanding of the economic environment. Regardless of your approach, it is important to distinguish between two cases: $m' \geq m_2^*$ and $m' < m_2^*$, where m_2^* is defined as the amount of money that allows a buyer to purchase q^* if she meets a type-2 seller.

d) Describe the equilibrium real balances, z, as a function of the nominal interest rate (on a perfectly safe yet perfectly illiquid bond), i.[6]

e) Based on your work in part (d), describe the equilibrium values q_1, q_2.

Finally, define the welfare function of this economy as the measure of the various KW market meetings times the net surplus generated in each meeting, i.e.,

$$\mathcal{W} = \sigma[u(q_2) - q_2] + (1 - \sigma)[u(q_1) - q_1].$$

Also, assume that the KW utility function is quadratic, i.e., $u(q) = (1 + \gamma)q - \frac{q^2}{2}$. (This will allow you to find closed form solutions for all the equilibrium objects.)

f) One would expect welfare to be increasing in the credit limit, C, i.e., $\partial\mathcal{W}/\partial C > 0$. Is this conjecture (always) correct in this model? Please explain intuitively.

[6] **Hint:** In the previous hint, I explained that the J function will behave differently around the point m_2^*. This property will also be reflected in the demand for real balances. More precisely, there will be a critical level of i, call it $\tilde{\imath}$, such that for $i \leq \tilde{\imath}$ buyers who meet type-2 sellers will be able to purchase q^*. What is that critical level $\tilde{\imath}$, and how does z behave on either side of it?

Question 3 (*25 points*)

In recent years there has been much discussion about the role of firm market power in macroeconomics. This question considers the economic consequences of a temporary (but persistent) rise in market power in the New Keynesian model.

There are a continuum of identical households and the representative household makes consumption (C) and labor supply (N) decisions to maximize lifetime expected utility. In linearized form, the household's Euler equation is:

$$E_t\hat{c}_{t+1} - \hat{c}_t = \frac{1}{\sigma}(\hat{i}_t - E_t\hat{\pi}_{t+1})$$

and their labor supply condition is given by:

$$\hat{w}_t = \sigma\hat{c}_t + \psi\hat{n}_t.$$

Monopolistically competitive intermediate goods firms produce an intermediate good of variety j using labor. Final goods firms purchase intermediate goods and transform them into a composite final good using a CES production function:

$$Y_t = \left(\int_0^1 y_t(j)^{\frac{\epsilon_t-1}{\epsilon_t}}dj\right)^{\frac{\epsilon_t}{\epsilon_t-1}}$$

The main difference from the baseline model is that there is now exogenous time variation in the degree of market power. Specifically, we assume that the elasticity of substitution between varieties, ϵ_t, is time varying. As a consequence, there will be some exogenous variation in the markup over time. Let's refer to this as a "markup shock", denoted by $\mu_t = \frac{\epsilon_t}{\epsilon_t-1}$.

In linearized form, the equilibrium conditions for firms are: $\hat{y}_t = \hat{n}_t$, $\hat{w}_t = \hat{mc}_t$, and

$$\hat{\pi}_t = \beta E_t(\hat{\pi}_{t+1}) + \lambda\hat{mc}_t + \lambda\hat{\mu}_t.$$

The resource constraint is: $\hat{y}_t = \hat{c}_t$. Monetary policy follows a simple Taylor Rule:

$$\hat{i}_t = \phi_\pi\hat{\pi}_t$$

The markup shock in percentage deviations from steady state, $\hat{\mu}_t$, evolves:

$$\hat{\mu}_t = \rho\hat{\mu}_{t-1} + e_t$$

where e_t is i.i.d. and $0 < \rho < 1$. In percentage deviations from steady state: $\hat{mc}_t$ is real marginal cost, $\hat{\mu}_t$ is the markup shock, $\hat{c}_t$ is consumption, $\hat{w}_t$ is the real wage, $\hat{n}_t$ is hours worked, $\hat{y}_t$ is output. In deviations from steady state: $\hat{i}_t$ is the nominal interest rate, $\hat{\pi}_t$ is inflation. λ is a function of model parameters, including the degree of price stickiness.[7] Assume that $\phi_\pi > 1$, $\sigma > 0$ and $\psi > 0$.

a) Using the relevant equations above, show that the natural rate of output in this model depends on the markup shock. In particular, show:

$$\hat{y}_t^n = -\frac{1}{\sigma + \psi}\hat{\mu}_t$$

b) Show that the Phillips Curve can be written as:

$$\hat{\pi}_t = \beta E_t(\hat{\pi}_{t+1}) + \lambda(\sigma + \psi)\hat{x}_t + \lambda\hat{\mu}_t$$

where $\hat{x}_t$ is the welfare relevant output gap, defined as $\hat{y}_t - \hat{y}_t^e$. (**Hint:** in this model the distortionary tax is the only stochastic element and $\hat{y}_t^e = 0$, which implies $\hat{x}_t = \hat{y}_t = \hat{c}_t$.)

c) Using the method of undetermined coefficients, find the response of the output gap, $\hat{x}_t$, and inflation, $\hat{\pi}_t$, to an exogenous increase in $\hat{\mu}_t$ when prices are sticky and monetary policy follows the Taylor Rule above. To do this, guess that the solution for each variable is a linear function of the shock $\hat{\mu}_t$.

d) Now suppose that, instead of following the simple Taylor Rule above, we choose $\hat{x}_t$ and $\hat{\pi}_t$ to maximize welfare. In particular, we will derive the optimal policy under **discretion**. Assuming that the steady state is efficient, a planner who cannot commit to future promises solves the following problem period by period:

$$\min_{\hat{\pi}_t, \hat{x}_t} \frac{1}{2}\left(\hat{\pi}_t^2 + \vartheta \hat{x}_t^2\right)$$

subject to

$$\hat{\pi}_t = \beta E_t(\hat{\pi}_{t+1}) + \kappa\hat{x}_t + \lambda\hat{\mu}_t, \tag{1}$$

taking $E_t(\hat{\pi}_{t+1})$ as given, and where $\kappa \equiv \lambda(\sigma + \psi)$. Using the first order conditions of this problem, show that the planner follows a *targeting rule* of the form

$$\hat{x}_t = -\frac{\kappa}{\vartheta}\hat{\pi}_t. \tag{2}$$

e) Finally, let's find the optimal policy. Combining equations (1) and (2) (and iterating forward), show that the inflation rate and the output gap are given by

$$\hat{\pi}_t = \frac{\lambda\vartheta}{\kappa^2 + \vartheta\left(1 - \beta\rho\right)}\hat{\mu}_t, \qquad \hat{x}_t = -\frac{\lambda\kappa}{\kappa^2 + \vartheta\left(1 - \beta\rho\right)}\hat{\mu}_t.$$

Is it possible to fully stabilize inflation and the output gap with optimal policy? *Briefly* explain.

[7] $\lambda = \frac{(1-\theta)(1-\beta\theta)}{\theta}$ where θ is the probability that a firm cannot adjust its price and $0 \leq \theta < 1$.

Question 4 (*25 points*)

This problem analyzes the welfare effects of a "capital injection" in a model with financial frictions. There are two periods, 0 and 1. There are two types of agents, consumers and entrepreneurs. Both types have a linear utility function, $c_0 + c_1$. Consumers have a large endowment of the consumption good in each period, and a unit endowment of labor in period 1 which they sell inelastically in a competitive labor market at the price w_1. Entrepreneurs have an endowment of the consumption good n_0 in period 0, and no initial capital. In period 0 they can borrow b_1 and invest k_1. In period 1 they hire workers at the wage w_1 and produce the consumption good according to the Cobb-Douglas production function

$$y_1 = k_1^{\alpha} l_1^{1-\alpha}.$$

Assume that k_1 completely depreciates after use in production.

Moreover, the entrepreneurs face the borrowing constraint

$$b_1 \leq \lambda(y_1 - w_1 l_1),$$

where $\lambda \in (0,1)$ is a given scalar. Assume the consumers endowment is large enough so that the gross interest rate is always 1 in equilibrium.

a) State the entrepreneurs' problem. Take the first order condition with respect to l_1. Show that the entrepreneurs will always choose l_1 to maximize profits in period 1, that is, they choose l_1 to solve

$$w_1 = (1-\alpha) k_1^{\alpha} l_1^{-\alpha}$$

Show that this implies that profits are a linear function of the capital stock k_1, that is,

$$y_1 - w_1 l_1 = R(w_1) k_1$$

for some function $R(\cdot)$ that depends only on w_1 and parameters.

b) Using the result from a), restate the entrepreneurs' problem as a simpler linear problem, replacing $y_1 - w_1 l_1$ by $R(w_1) k_1$. Characterize the solution to the entrepreneurs' problem with first order conditions.

c) Show that if $\lambda R(w_1) < 1 \leq R(w_1)$ the entrepreneurs' investment in capital is strictly positive and finite. What happens if $\lambda R(w_1) \geq 1$? What if $R(w_1) < 1$?

d) Show that if the entrepreneurs are constrained, their choice of capital is

$$k_1 = \frac{n_0}{1 - \lambda R(w_1)}$$

e) Show that the equilibrium level of capital solves

$$k_1 = \frac{n_0}{1 - \lambda \frac{\alpha}{k_1^{1-\alpha}}}$$

Show that k_1 and the entrepreneurs' profits are increasing in n_0. **Hint:** Remember that we limit attention to the case in which $\lambda R(w_1) < 1 \leq R(w_1)$.

f) Assume that the entrepreneurs are constrained. Suppose the government taxes a lump-sum τ to consumers in period 0, and transfers the receipts from the tax to the entrepreneurs. Derive an expression for the utility of consumers and entrepreneurs as a function of τ.

g) Show that if n_0 is sufficiently small, a small positive tax increases the utility of both consumers and entrepreneurs.

Question 5 (*25 points*)

Assume there is a continuum of measure 1 of individuals, each endowed with one unit of a storable consumption good. There are three time periods, $t = 0, 1, 2$. At $t = 0$ individuals have two options with regards to how they can invest their endowment. They can either stuff it in their mattress, where it gets a gross return equal to 1 (i.e., $1+r = 1$), or they can invest it in a long-term project that yields a gross return $R = 4$ in period two. For example, an individual that invests an amount I will receive $4I$ in period two, and has $1 - I$ stuffed under the mattress. In $t = 1$, individuals have the option of liquidating the long-term project at a penalty. If they liquidate, they only receive a return $L = 1/4$ in period 1, rather than the return $R = 4$ in period 2.

At time $t = 1$, a fraction $\pi = 1/2$ of the individuals receive a liquidity shock. These individuals are "impatient" and only value consumption in period one. The fraction $1 - \pi$ individuals that do not receive a liquidity shock are "patient" and only value consumption in period two. At time $t = 0$, all individuals have the same chance of being hit by the liquidity shock. Assume that individuals do not discount the future, so that their ex-ante expected utility is given by

$$U = \pi u(c_1) + (1 - \pi)u(c_2),$$

where c_1 and c_2 are the consumption in period 1 and 2, respectively, and $u(c) = -1/c$.

a) Assume there are no financial markets available, so that individuals must simply invest on their own. Given that an individual has invested an amount I at time $t = 0$, what will be the optimal levels of consumption, c_1, c_2, if:

(i) the individual receives a liquidity shock (i.e. is impatient)

(ii) the individual does not receive a liquidity shock (i.e. is patient)

b) What is the optimal level of investment, I^*? What is the ex-ante expected utility of an individual?

c) Now suppose that when types are realized in period 1, this information is publicly observable. Suppose there exists a social planner that individuals entrust all of their endowment to at time 0. The social planner will pay impatient individuals $\hat{c}_1$ in period 1 and patient individuals $\hat{c}_2$ in period 2.

(i) Solving the social planner's problem, what is $\hat{c}_1$ and $\hat{c}_2$? How much does the social planner invest? That is, what is $\hat{I}$?

(ii) What is an individual's ex-ante expected utility now?

(iii) Why can the planner improve over autarky? How would your answer change if $L = 1$?

d) Now suppose an agent's type is private information, and the social planner can only offer a contract contingent on an individual's announcement of her type at time 1. Furthermore, at time 1, she meets each agent once with the meeting order randomly determined. If individuals report honestly, can the social planner offer the same contract as in question c)? Is it optimal for an individual to report honestly when everyone else does? Explain in 1-2 sentences how this planner can be interpreted as a bank.

e) Suppose all agents fear a bank run, and each agent reports to the bank at time 1 as being impatient.

(i) How many individuals will get paid by the bank before it runs out of money in period 1? Given this, explain why a bank run can be an equilibrium, i.e., why it is optimal for a "patient" individual to run on the bank when she expects a bank run.

(ii) Suppose the bank implements a policy of only paying the first π individuals that show up at time 1, and the rest will get paid at time 2 (i.e. it suspends convertibility). Will this eliminate the bank run as an equilibrium? Can you think of any problems with this policy?